

RESEARCH ARTICLE OPEN ACCESS

Beyond Climate Targets: Exploring When and How Female Directors Influence Corporate Decarbonization Transparency

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Received: 4 November 2024 | **Revised:** 15 January 2025 | **Accepted:** 22 January 2025

Funding: Open access publishing facilitated by the Universidad de Salamanca, as part of the Crue Spanish Universities-CSIC Alliance.

Keywords: board gender diversity | climate change | climate transparency | critical mass | decarbonization strategies | external pressures | institutional pressures

ABSTRACT

The 2015 Paris Agreement established an international commitment to limit global warming to 1.5°C, which requires climate neutrality through deep cuts in greenhouse gas emissions. In pursuit of this goal, companies worldwide are adopting decarbonization strategies that are increasingly aligned with principles of transparency and accountability. This study examines a sample of 6575 large global companies to analyze the impact of board gender diversity on climate-related disclosures. Our findings show that the presence of at least one female director increases corporate transparency regarding decarbonization targets, timelines, strategic levers, and performance metrics. Thus, this study challenges the critical mass theory by demonstrating that even a single female director adds unique value in promoting sustainability transparency. Furthermore, we show that contextual factors—such as industry environmental sensitivity, regional regulatory frameworks and climate-related business opportunities—moderate the influence of female directors on decarbonization transparency. These findings advance corporate governance and sustainability research by providing a multidimensional understanding of how board gender diversity drives transparency, particularly in sustainability-sensitive industries and regulatory environments. On a practical level, the findings highlight the strategic value of gender-diverse boards for managers, investors, and policymakers seeking to enhance corporate accountability and align with global sustainability goals. By underscoring the transformative role of female directors in promoting transparent and responsible corporate practices, this research contributes actionable insights to the transition to a net-zero economy.

1 | Introduction

Climate change is one of the most urgent challenges of our time, with significant environmental and economic implications worldwide (Ararat and Sayedy 2019; Ooi et al. 2019; Al-Najjar and Salama 2022). This phenomenon, characterized by rising global temperatures, increased climate variability, and the intensification of extreme weather events, presents serious risks

for businesses, which are collectively responsible for over 70% of global greenhouse gas (GHG) emissions (Morrison, Jia, and Arora 2024). As major players in the global economy, businesses hold both the capability and responsibility to adopt strategies that contribute to mitigating climate change and achieving the goals outlined in the Paris Agreement (Qureshi et al. 2020; Fan, Tang, and Pan 2021; Al-Najjar and Salama 2022; Wang et al. 2024). Within this context, corporate climate reporting has

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become crucial for stakeholders' decision-making (Al-Qahtani and Elgharbawy 2020; Dutta and Dutta 2021; Ngo et al. 2023), particularly for investors (Amran, Periasamy, and Zulkafli 2014; Ilhan et al. 2023; Subedi and Zoet 2024; Vestrelli, Colladon, and Pisello 2024). Companies are increasingly called upon not only to define specific emission reduction goals but also to disclose detailed data on targets, timelines, strategies, and progress towards a low-carbon economy (Bauckloh et al. 2023). Climate disclosure is thus emerging as a critical practice for enhancing transparency, promoting responsible behavior, and appropriately managing climate risks, helping companies maintain stakeholder trust (Datt et al. 2022; Guo, Zhao, and Yang 2022; Abbasi et al. 2024).

The board of directors, as the apex of corporate governance, plays a pivotal role in both the development and oversight of sustainability strategies (Hollindale et al. 2019; Ooi et al. 2019). Decisions regarding the adoption of effective climate policies and the transparency of these policies to stakeholders depend substantially on board structure and composition (Amran, Periasamy, and Zulkafli 2014; Ararat and Sayedy 2019). In this sense, research increasingly highlights that board diversity broadens the range of skills, knowledge, and perspectives within the board (Nielsen and Huse 2010) and enhances the integration of environmental, social, and governance (ESG) criteria in corporate strategy (Al-Qahtani and Elgharbawy 2020; Wasiuzzaman and Subramaniam 2023) and disclosure practices (Ibrahim and Hanefah 2016; Nicolò et al. 2022). In particular, board gender diversity (BGD) has been recognized as a key attribute for comprehending the risks and opportunities associated with climate change (Ben-Amar, Chang, and McIlkenny 2017; Nuber and Velte 2021; Al-Najjar and Salama 2022; García-Sánchez et al. 2023b). Evidence suggests that women in leadership roles display heightened sensitivity to environmental and social concerns (McGuinness, Vieito, and Wang 2017; Tingbani et al. 2020; Monteiro, García-Sánchez, and Aibar-Guzmán 2022), reflected in a proactive approach to stakeholder expectations for transparency and accountability (Al-Shaer and Zaman 2016; Ben-Amar, Chang, and McIlkenny 2017). On boards, female directors are often associated with fostering a responsible corporate culture, advancing sustainable and decarbonization practices (Nielsen and Huse 2010; Ciocirlan and Pettersson 2012; Atif et al. 2021; García-Sánchez et al. 2023b) and encouraging clear, precise disclosure regarding progress in these areas (Liao, Luo, and Tang 2015; Hossain et al. 2017; Hollindale et al. 2019).

Although previous research has examined the influence of female directors on corporate climate disclosure, most studies focus primarily on GHG disclosure (Prado-Lorenzo and Garcia-Sanchez 2010; Liao, Luo, and Tang 2015; Ben-Amar, Chang, and McIlkenny 2017; Hollindale et al. 2019; Al-Qahtani and Elgharbawy 2020; Tingbani et al. 2020). The few studies that examine broader climate-related coverage are often limited to a single country (e.g., Ararat and Sayedy (2019): Turkey; Charumathi and Rahman (2019): India). Thus, the influence of BGD on the disclosure of information about firms' decarbonization strategies remains underexplored. Moreover, recent studies have shown that the COVID-19 pandemic has led to shifts in BGD, which may affect the influence of female directors on corporate strategies (García-Sánchez et al. 2024b). Meanwhile, contextual

factors have been shown not only to influence corporate sustainability reporting (Nicolò and Andrades-Peña 2024) but also to moderate female directors' influence on such disclosure practices (Byron and Post 2016; Nadeem, Zaman, and Saleem 2017; Nicolò et al. 2022; Wasiuzzaman and Subramaniam 2023).

This study aims to bridge existing gaps by addressing two key challenges in corporate governance: gender diversity on boards and transparency in climate change disclosures (Ben-Amar, Chang, and McIlkenny 2017; Al-Qahtani and Elgharbawy 2020). Specifically, we analyze the effect of BGD on the transparency of decarbonization strategy disclosures, focusing on whether the presence of female directors enhances the breath and clarity of climate-related reporting. Additionally, we explore how diverse organizational and institutional contexts may moderate the impact of female directors in this area. Thus, our analysis offers a nuanced perspective on corporate climate commitments and transparency in advancing decarbonization goals, shedding light on the specific role of female directors and the factors that shape their impact.

Drawing on an unbalanced panel dataset of 6575 firms (46,561 observations) over the 2015–2022 period, our findings support the hypothesis that increased BGD is positively associated with greater transparency in decarbonization disclosures. We observe that the presence of even a single woman on the board is associated with more comprehensive reporting on decarbonization targets, timelines, strategies, and performance. Moreover, female directors contribute to the quality of these disclosures, ensuring that firms provide clear and detailed information on their decarbonization initiatives. Additionally, our results reveal that contextual factors—such as industry environmental sensitivity, regional regulatory frameworks, particularly the institutional setting of the European Union (EU), and climate-related business opportunities—moderate the influence of female directors on corporate decarbonization transparency. Specifically, although industry environmental sensitivity amplifies their impact, both the EU institutional environment and climate-related business opportunities slightly attenuate it.

This study makes several key contributions to the corporate sustainability and governance literature by offering a multidimensional understanding of the role of female directors in promoting transparency around decarbonization strategies. Unlike prior research, our findings indicate that the positive influence of female directors on transparency does not require a critical mass of women on the board; each female director appears to bring distinct value that strengthens the organization's commitment to sustainability transparency. This insight challenges critical mass theory and suggests new directions for examining the influence dynamics of boards with low female representation in ESG contexts.

We also contribute to institutional theory by demonstrating the moderating role of contextual factors on the influence of female directors on sustainability disclosure practices. By providing empirical evidence on how industry environmental sensitivity and EU regulations affect the relationship between gender diversity and transparency of decarbonization-related information, we broaden the understanding of how institutional contexts and business opportunities shape corporate sustainability

practices. Specifically, this research extends institutional theory by showing that environments demanding sustainability and transparency increase the value of BGD, except when institutional mandates reduce the reliance on specific board compositions. Finally, this research contributes to discussions on gender perspectives in the transition to a sustainable, zero-carbon economy by highlighting the role of female directors in promoting transparent and responsible corporate practices.

The remainder of the paper is organized as follows: Section 2 outlines the theoretical framework and hypothesis development. Section 3 describes the study's empirical approach. Section 4 presents and discusses the main findings. The final section summarizes the study's conclusions and implications.

2 | Background and Hypothesis Development

2.1 | Theoretical Framework

The influence of BGD on the adoption and disclosure of ESG strategies has been examined through various theoretical frameworks, including agency theory, resource dependence theory, upper echelon theory, and stakeholder theory (Nguyen, Ntim, and Malagila 2020; Amorelli and García-Sánchez 2021). Nevertheless, several scholars (e.g., Elmagrhi et al. 2019; García-Sánchez et al. 2023b) argue that the role of female directors in this domain cannot be fully explained by any single theory, advocating for a combination of theoretical approaches. Following this recommendation and in alignment with recent studies (e.g., Al-Najjar and Salama 2022; Nicolò et al. 2022; Abbasi et al. 2024), this study adopts a multitheoretical framework to explain the influence of BGD on the transparency about decarbonization strategies. Specifically, we apply resource dependence theory, critical mass theory, and institutional theory.

Resource dependence theory posits that firms are open systems that depend on external resources that are often controlled by other actors in the environment (Pfeffer and Salancik 1978; Hillman and Dalziel 2003). To mitigate uncertainty and manage dependencies, firms strategically assemble their boards to enhance their access to critical resources, including diverse perspectives, knowledge, and networks (Pfeffer and Salancik 1978), and to address social and environmental challenges (Manita et al. 2018; Nicolò et al. 2022; García-Sánchez et al. 2024d). From this perspective, board capital refers to the value that board members bring to the organization in terms of experience, knowledge, skills, and social networks (Zahra and Pearce 1989). Female directors enhance board capital (Charumathi and Rahman 2019; Hollindale et al. 2019; Orazalin and Baydauletov 2020) by bringing diverse skills and perspectives that complement those of male directors (Ben-Amar, Chang, and McIlkenny 2017; Nguyen, Ntim, and Malagila 2020; Amorelli and García-Sánchez 2021; Nicolò et al. 2022).

According to critical mass theory, a threshold or “critical mass” of participation must be reached for minorities within a group to exert meaningful influence on collective decision-making (Kanter 1977). Originally applied to the study of women's representation in politics, this theory has been extended to the corporate arena, suggesting that a sufficient

number of female directors is required to meaningfully impact board decision-making (Nicolò et al. 2022; Wasiuzzaman and Subramaniam 2023). Thus, low representation of women on boards may limit their impact on policies and decisions, whereas reaching a critical threshold allows female directors to drive substantive change (Torchia, Calabrò, and Huse 2011; Joecks, Pull, and Vetter 2013; Atif, Liu, and Huang 2019). Kristie (2011, 22) illustrates this concept as follows: “One is a token, two is presence, and three is voice.” By shifting the balance of influence within the boardroom, a critical mass of female directors amplifies their collective voice, enabling more consistent advocacy for robust and transparent sustainability practices (Post, Rahman, and Rubow 2011; Torchia, Calabrò, and Huse 2011; Charumathi and Rahman 2019; Atif et al. 2021).

Finally, institutional theory posits that organizations adapt to societal norms and expectations in order to maintain legitimacy (DiMaggio and Powell 1983). Thus, institutional pressures influence sustainability disclosure practices, although this influence may vary depending on the nature of the institutional pressures faced by firms (Nicolò et al. 2024). According to institutional theory, these pressures can be understood as coming from three main sources (DiMaggio and Powell 1983): coercive pressures from regulatory requirements or powerful stakeholders, mimetic pressures from industry leaders setting precedents in climate transparency, and normative pressures from societal expectations of sustainability. These forces shape how companies navigate and respond to environmental issues (Nicolò et al. 2024). From this perspective, institutional pressures may shape board decisions on climate transparency, amplifying the impact of gender-diverse boards.

2.2 | Literature Review and Hypothesis Development

Research on the drivers of climate change disclosure has identified both internal factors, such as firm characteristics and governance mechanisms, and external factors as key determinants (Al-Qahtani and Elgharbawy 2020; Song and Xian 2024). Among the internal factors, researchers show that larger and more profitable firms, those with larger and more specialized boards, a higher presence of institutional investors, and those in fossil fuel-intensive sectors are more likely to disclose climate-related information (Caby, Ziane, and Lamarque 2020; Subedi and Zoet 2024). External factors, including a country's economic development, legal framework, cultural dimensions, and attitudes towards climate change, serve as sources of institutional pressure that positively influence disclosure practices (Velte, Stawinoga, and Lueg 2020; Bose et al. 2024). Moreover, several studies highlight that many of these factors influence not only the likelihood of disclosure but also the quality and materiality of the information disclosed (Mateo-Márquez, González-González, and Zamora-Ramírez 2022; Rodríguez, Gambetta, and García-Benau 2023).

Previous studies have examined the relationship between BGD and corporate transparency on sustainability matters (e.g., Post, Rahman, and Rubow 2011; Frias-Aceituno, Rodríguez-Ariza, and García-Sánchez 2013; Al-Shaer and Zaman 2016; Ibrahim and Hanefah 2016; Rao and Tilt 2016;

Elmagrhi et al. 2019; García-Sánchez et al. 2024a). In general, these studies document a positive association between the presence of female directors and environmental disclosure (Qureshi et al. 2020; Nicolò et al. 2022; Wasiuzzaman and Subramaniam 2023). Specifically, in the context of climate-related information, empirical evidence highlights a similar positive effect of BGD on voluntary climate disclosures. Liao, Luo, and Tang (2015), Al-Qahtani and Elgharbawy (2020), and Tingbani et al. (2020) document a positive relationship between the proportion of female directors and the likelihood of large UK firms disclosing information on GHG and the comprehensiveness of such disclosures. Similar findings are reported by Ben-Amar, Chang, and McIlkenny (2017) and Hollindale et al. (2019) in Canada and Australia, respectively. Charumathi and Rahman (2019) and Ooi et al. (2019) also report a positive association between BGD and carbon disclosure in emerging markets (India and Malaysia, respectively). However, some studies offer contrasting perspectives: Amran, Periasamy, and Zulkafli (2014) find a negative relationship between BGD and climate disclosure in a sample of 13 Asia-Pacific countries, Prado-Lorenzo and Garcia-Sanchez (2010) report no significant association in an international sample, and Ararat and Sayedy (2019) note that, in Turkish firms, female presence on boards does not promote GHG disclosure, although it does so in board committees. Overall, this empirical evidence suggests that gender-diverse boards are more likely to advocate for the disclosure of information on decarbonization strategies.

Because of their different socialization and professional experiences, female directors are often perceived as agents of change, advocating for greater accountability and stakeholder engagement (Nicolò et al. 2022). Their presence can lead to increased sensitivity to external pressures (Harjoto, Laksmana, and Lee 2015; Al-Shaer and Zaman 2016; Rao and Tilt 2016; Manita et al. 2018), such as investor demands and societal expectations for climate action (Ben-Amar, Chang, and McIlkenny 2017; Ilhan et al. 2023; García-Sánchez et al. 2023b; Subedi and Zoet 2024). This alignment with broader stakeholder concerns fosters proactive transparency practices (Ben-Amar, Chang, and McIlkenny 2017; Hollindale et al. 2019; Al-Qahtani and Elgharbawy 2020; Wasiuzzaman and Subramaniam 2023; Abbasi et al. 2024; Nicolò and Andrades-Peña 2024; García-Sánchez et al. 2024a), especially in areas such as decarbonization, where public scrutiny and regulatory demands are increasing (Amran, Periasamy, and Zulkafli 2014; Bauckloh et al. 2023; Ngo et al. 2023). Consequently, from the perspective of resource dependence theory, BGD can be seen as a strategic response that strengthens a firm's strategic capabilities and aligns resource acquisition and use with evolving sustainability imperatives (Nicolò et al. 2022), thereby enhancing the board's ability to address complex issues such as climate change and decarbonization and effectively respond to stakeholder demands through climate-related disclosures (Ben-Amar, Chang, and McIlkenny 2017; Ooi et al. 2019; Al-Najjar and Salama 2022; García-Sánchez et al. 2023b; Abbasi et al. 2024).

In addition to bringing human and relational capital to the board, which favors different perspectives, female directors also bring different leadership styles (Rao and Tilt 2016; Nicolò et al. 2022).

Bass (1999, 10–11) proposed a typology of leadership styles, distinguishing between transactional and transformational leadership. The former refers to “the exchange relationship between the leader and followers to satisfy their own interests,” whereas the latter describes transformational leaders as those who can mobilize their followers “beyond their immediate self-interests” and motivate them to achieve broader societal goals. From this perspective, female and male directors tend to differ in their leadership styles (Monteiro, García-Sánchez, and Aibar-Guzmán 2022; Nicolò and Andrades-Peña 2024), with the former more likely to adopt a transformational leadership style and the latter more likely to adopt a transactional leadership style (Reuvers et al. 2008; Stempel, Rigotti, and Mohr 2015). Thus, given that climate change disclosures that go beyond legal requirements involve “transformational change” in transparency policies, they would be encouraged by transformational leadership from directors, and thus, greater BGD would increase the transparency of decarbonization-related information.

Therefore, considering that female directors not only add value in terms of capital and resources but also promote more ethical management and a commitment to transparency on environmental and social issues, the following hypothesis is proposed:

Hypothesis 1. *Female presence on boards is associated with greater transparency of decarbonization-related information.*

Although most studies suggest that the presence of female directors increases corporate transparency on environmental and social issues (Al-Qahtani and Elgharbawy 2020; Qureshi et al. 2020; Nicolò et al. 2022; Wasiuzzaman and Subramaniam 2023; García-Sánchez et al. 2024a), it may be necessary for women to hold a sufficient number of seats to be more than a token presence (Torchia, Calabrò, and Huse 2011; Atif et al. 2021). From the perspective of critical mass theory, isolated female directors may face challenges such as tokenism or resistance to their contributions, which may dilute their ability to shape organizational priorities, including transparency in environmental disclosure (Manita et al. 2018; De Masi et al. 2021). However, when women's representation reaches a critical mass, it fosters a supportive environment for their perspectives and initiatives to gain traction (Post, Rahman, and Rubow 2011; Charumathi and Rahman 2019). This transformation strengthens the board's ability to prioritize sustainability concerns and respond to external demands for greater decarbonization transparency. In the context of sustainability disclosure, authors such as Manita et al. (2018) and De Masi et al. (2021) show that a critical mass of female directors is required to significantly impact disclosure, whereas Ben-Amar, Chang, and McIlkenny (2017), Charumathi and Rahman (2019), Hollindale et al. (2019), and Abbasi et al. (2024) do the same with respect to climate disclosure.

Therefore, considering the possibility that a critical mass of female directors is required for their perspectives on corporate transparency with respect to decarbonization strategies to be influential, we hypothesize the following:

Hypothesis 2. *Female board presence is associated with greater transparency of decarbonization-related information only when it reaches a critical mass.*

Previous research suggests that external pressures are instrumental in driving sustainability reporting (García-Sánchez et al. 2023a; Nicolò and Andrades-Peña 2024), including climate-related disclosures (Luo, Lan, and Tang 2012; Amran, Periasamy, and Zulkafli 2014; Kalu, Buang, and Aliagha 2016; He et al. 2019). Moreover, in addition to boardroom power dynamics, research has also shown that the influence of female directors on firm strategy and corporate transparency is also influenced by the characteristics of the organizational and institutional context in which firms operate (Byron and Post 2016; Ararat and Sayedy 2019; Sarhan, Ntim, and Al-Najjar 2019; Wasiuzzaman and Subramaniam 2023; Almarayeh, Aibar-Guzman, and Suárez-Fernández 2024). From the perspective of institutional theory, the broader institutional environment shapes the opportunities and constraints that female directors encounter in advancing the sustainability agenda (Nicolò et al. 2024). Therefore, female directors are more likely to succeed in increasing decarbonization transparency when institutional pressures from the organizational and institutional context align with their advocacy efforts. This alignment amplifies their ability to influence decision-making processes and promote transparency and collectively shape corporate behavior and decision-making processes (Wasiuzzaman and Subramaniam 2023; García-Sánchez et al. 2024b).

Industries characterized by high environmental sensitivity, such as mining, transportation, and oil and gas, face increased scrutiny from stakeholders, which incentivizes firms to adopt more transparent practices (Qureshi et al. 2020). In addition, firms operating in environmentally sensitive sectors must comply with sustainability and transparency standards (Monteiro and Aibar-Guzmán 2010), forcing them to increase transparency to reflect their decarbonization strategies. These coercive pressures create an enabling environment for female directors to influence transparency, as alignment with sustainability norms not only satisfies external demands but also enhances the social legitimacy of the firm. In this sense, in firms operating in environmentally sensitive sectors, female

directors may find stronger organizational support for sustainability initiatives as they align with industry-level norms and stakeholder expectations.

In addition, regulatory frameworks create coercive pressures that force companies to disclose extensive information (Wasiuzzaman and Subramaniam 2023; Nicolò et al. 2024). The EU, in particular, is recognized as a leading region in promoting sustainable development and corporate transparency (Qureshi et al. 2020; Nicolò et al. 2022). The EU's stringent ESG regulations create an environment in which robust transparency practices are prioritized, leading boards of directors to adopt proactive transparency approaches (Ferrer, López-Arceiz, and del Rio 2020; Lippai-Makra, Kovács, and Kiss 2022; García-Sánchez et al. 2023a; Nicolò and Andrades-Peña 2024). In this sense, the stringent sustainability disclosure requirements mandated by European directives, such as the Corporate Sustainability Reporting Directive (CSRD) and the European Sustainability Reporting Standards (ESRS), force companies to disclose detailed information related to decarbonization. Such coercive pressures reduce internal resistance to transparency initiatives, allowing female directors to use regulatory compliance as a platform for advancing decarbonization disclosure practices.

Finally, firm-level opportunities related to climate change, such as the development of green technologies or access to sustainable financing, further moderate the relationship between BGD and transparency. Firms that perceive climate change as a strategic opportunity are more likely to integrate sustainability into their core operations and disclosures as a means of gaining competitive advantage (Aibar-Guzmán et al. 2024; García-Sánchez et al. 2024d). The mimetic pressures that such actions exert on other firms reinforce the advocacy efforts of female directors, who can champion decarbonization transparency as a means of enhancing the firm's market positioning and adaptive capacity in a low-carbon economy.

Thus, the following overarching hypothesis is proposed:

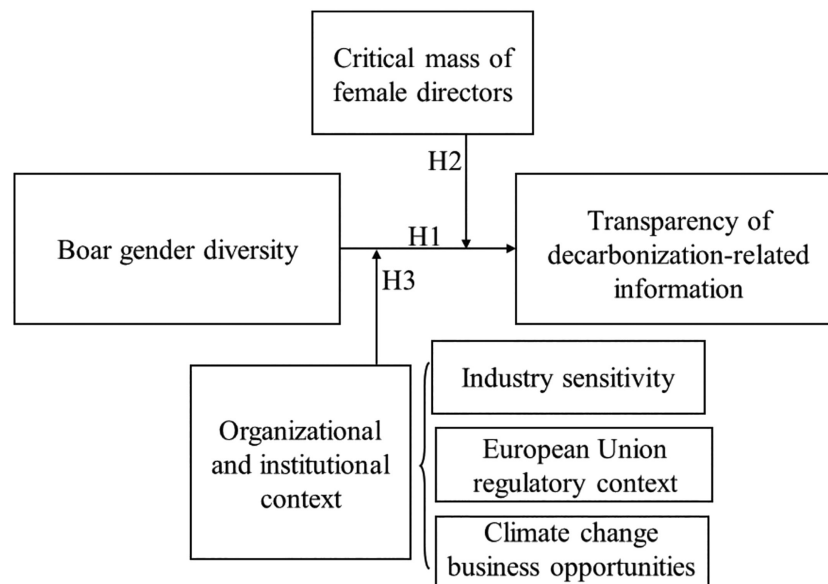


FIGURE 1 | Research model.

Hypothesis 3. *The influence of BGD on the transparency of decarbonization-related information is positively moderated by the organizational and institutional contexts in which firms operate.*

To further delineate this hypothesis, three subhypotheses are presented:

Hypothesis 3a. *The influence of BGD on the transparency of decarbonization-related information is positively moderated by the environmental sensitivity of the industry to which a firm belongs.*

Hypothesis 3b. *The influence of BGD on the transparency of decarbonization-related information is positively moderated by the regulatory context of the EU.*

Hypothesis 3c. *The influence of BGD on the transparency of decarbonization-related information is positively moderated by the firm's climate change-related business opportunities.*

Figure 1 illustrates the overall research model.

3 | Methodology

3.1 | Sample

The empirical testing of our research hypothesis requires a rigorous methodological design to ensure the logical soundness, adequacy, and representativeness of the data. The first element to be defined is the population of interest. In alignment with prior research (Aibar-Guzmán et al. 2023, 2024), we have selected large global companies, as these entities are not only subject to greater scrutiny due to their visibility, but they also possess the resources and capabilities to implement ambitious climate change initiatives (García-Sánchez et al. 2023b, 2024d).

Drawing on data from the Refinitiv database, the final sample comprises an unbalanced panel dataset with 6575 companies

(46,561 observations) covering the period of 2015–2022. The starting year reflects the signing of the Paris Agreement, marking an intensification in corporate commitments to climate action. Table 1 summarizes the sample selection stages and criteria, all defined by the availability of essential information.

3.2 | Variables

In order to analyze our research hypothesis, we have defined a set of variables according to the previous literature and the main organizations in the field. We have classified them into dependent, independent, moderating, and control variables according to their role in the empirical part.

In this sense, the dependent variable, DescInf, corresponds to a composite indicator ranging from 0 to 100, capturing the transparency of companies regarding their decarbonization commitments. This indicator is calculated as $\sum_{i=1}^9 \frac{\text{DescItem}_i}{9}$, based on nine items that reflect disclosures related to decarbonization agreements and commitments, implemented measures, and progress made in emission reduction. These items were selected according to various standards and guidelines, such as Regulation (EU) 2018/1999 of the European Parliament and the Council on the Governance of the Energy Union and Climate Action and the Task Force on Climate-related Financial Disclosures (TCDF) reports. As for the procedure to create the DescInf score, we identified the information available in the Refinitiv database for the nine items in Panel A of Table 2. The items correspond to dummy variables that take the value of 1 if the company disclosed information on that item and 0 otherwise.

In addition, to ensure robust results, we created a categorical variable Omission, which takes the value 1 if a company omits information items DI8 and DI9 and 0 otherwise.

Consistent with prior studies, our independent variable, BGD, is proxied by the proportion of women on the board (Prado-Lorenzo and Garcia-Sanchez 2010; Liao, Luo, and Tang 2015; Charumathi and Rahman 2019; Tingbani et al. 2020; Al-Najjar and Salama 2022; García-Sánchez et al. 2023b, 2024a), coded here as the variable FemD.

For the critical mass variable, we use different approaches. Model 2a relates female representation to the three types of boards defined by Seebeck and Vetter (2022) and García-Sánchez et al. (2023b) based on gender composition: (i) Skewed boards, where women hold less than 20% of seats and men dominate; (ii) Tilted boards, with 20%–40% female representation but no gender balance; and (iii) Balanced boards, with 40%–60% female presence. These categories are operationalized by three dichotomous variables (Skewed, Tilted, and Balanced), which take the value 1 if the proportion of female directors falls within each range and 0 otherwise.

Inspired by Atif et al. (2021), Model 2b includes dummy variables for boards with one female director (OneWomen), two female directors (TwoWomen), and three or more female directors (3+ Women) that take the value of 1 when the proportion of female directors falls within each count.

TABLE 1 | Stages and criteria for sample selection.

Source	Refinitiv
Stages and criteria	Number of companies
<i>Criterion 1.</i> Companies with information on the presence/absence of a public commitment to decarbonization. Companies with missing values for this variable are excluded	8905
<i>Criterion 2.</i> Availability of the necessary information for the rest of the variables that make up the empirical models	7578
<i>Criterion 3.</i> Availability of the required information per company for at least 5 consecutive years	6575

TABLE 2 | Variables.

Panel A. Information items on business decarbonization			
			Frequency (%)
DI1. The company have a public compromise with emission reduction			60.80
DI2. The company has a set targets or objectives to be achieved on emission reduction (i.e., percentage of emission reduction target)			55.18
DI3. The company has fixed the year by which the emission reduction target is set			34.42
DI4. The company has established different scenarios to ensure the resilience of the strategy (information on changes in the strategy to address possible climate changes)			23.09
DI5. The company reports on the decarbonization levers and the key actions and tools planned to achieve the established objectives within the indicated period			52.29
DI6. The company has the organizational structures related to sustainability responsibilities			41.14
DI7. The company has approved that the board and/or committees consider issues related to sustainability and climate when reviewing and guiding the strategy, action plans, etc.			52.60
DI8. The company presents information on its climate coal			39.61
DI9. The company report the percentage of target annual reduction in emissions			32.27
Panel B. Independent and control variables			
		Mean	Std. dev.
DescInf	Composite index of decarbonization information disclosures	43.49	26.52
FemD	Proportion of female directors	19.17	14.08
EnvPerf	Environmental score from Refinitiv	46.34	20.89
Board	Number of directors	9.62	3.28
Meet	Number of meetings of the board	9.44	5.49
InD	Proportion of independent directors	61.64	24.95
Size	Natural logarithm of total assets	22.17	1.97
Lev	Debt to total assets ratio	18.59	19.68
ROA	Return on assets	5.43	7.15
Investment	Investment to sales ratio	19.36	1.91
Investors	Proportion of shares held by institutional investors	8.86	6.82
		Frequency (%)	
SustInd		19.73	
EU		12.39	
Oppo		16.06	
Duality	Dummy takes value 1 if the CEO is also the Chair of the board	34.39	
SustCom	Dummy takes value 1 if firms have a sustainability committee	52.69	
DisrupEvents	Dummy takes value 1 for 2020 to 2022	40.42	

In Model 2c, we introduced another measure with the variable NoWomen, a dummy that identifies all-male boards. The variable takes the value 1 if the directors are all male and 0 otherwise. Finally, in Models 2d and 2e, we include the presence of female directors and the square of their number and proportion, respectively.

Regarding the analysis of the effect moderating contexts proposed by Hypothesis 3, the term Scenario refers to

three different variables: SustInd, EU, and Oppo to test Hypotheses 3a, 3b, and 3c, respectively. SustInd captures industry-specific environmental sensitivity. It takes a value of 1 for sectors with high environmental impact, such as mining and quarrying, road transport, and oil and gas, and 0 for other sectors. The institutional environment of the EU, where sustainability and transparency are prioritized, is proxied by the variable EU, a dummy that takes a value of 1 for companies

domiciled in EU member states and 0 otherwise. Lastly, the variable *Oppo* denotes the presence of business opportunities linked to decarbonization, coded as 1 for firms that report business opportunities related to emission reduction and 0 otherwise. The interaction terms between these variables and BGD (*FemD*SustInd*, *FemD*EU*, and *FemD*Oppo*) capture the moderating effects of industry-specific environmental sensitivity, the EU institutional environment, and the existence of decarbonization-related business opportunities on the influence of BDG on the transparency of decarbonization strategies, respectively.

To minimize potential biases, we include several control variables in the empirical model. The conceptualization of these variables is detailed in Panel B of Table 2. For instance, the variable *EnvPerf* controls for corporate environmental performance, as prior research has shown that better-performing companies tend to disclose more relevant information (i.e., Clarkson et al. 2008; Prado-Lorenzo and Garcia-Sanchez 2010). Following previous studies (Ben-Amar, Chang, and McIlkenny 2017; Al-Qahtani and Elgharabawy 2020; Atif et al. 2021; Al-Najjar and Salama 2022; García-Sánchez et al. 2023a, 2024a; Abbasi et al. 2024), board effectiveness is represented by the variables *BoarD*, *Meet*, and *InD*, which capture board size, activity, and independence, respectively. Additionally, *Duality* and *SustCom* control for CEO duality and the existence of a sustainability-focused committee. To represent firm-specific capabilities and resources, we include the variables *Size*, *Lev*, *ROA*, and *Invest*, reflecting firm size, debt level, profitability, and investment, respectively. These variables' relevance to corporate transparency policies is established in previous literature (Daradkeh et al. 2023; Wang et al. 2024). Furthermore, we control for strategic investors, given their vested interest in climate-related risks (Aibar-Guzmán et al. 2023; Ilhan et al. 2023), as well as for the regulatory impacts of recent events like the COVID-19 pandemic and the war in Ukraine (Ben-Amar, Comyns, and Martinez 2023; García-Sánchez et al. 2024c).

3.3 | Empirical Model

Based on the previous sections and the literature, we define different dependence models to test our hypothesis. In this sense, Model 1 below is designed to test our first research hypothesis (Hypothesis 1):

$$\text{DescInf}_{it} = \delta_0 + \delta_1 \text{FemD}_{it} + \delta_2 \text{EnvPerf}_{it} + \delta_3 \text{BoarD}_{it} + \delta_4 \text{Meet}_{it} + \delta_5 \text{InD}_{it} + \delta_6 \text{Duality}_{it} + \delta_7 \text{SustCom}_{it} + \delta_8 \text{Size}_{it} + \delta_9 \text{Lev}_{it} + \delta_{10} \text{ROA}_{it} + \delta_{11} \text{Investment}_{it} + \delta_{12} \text{Investors}_{it} + \delta_{13} \text{DisrupEvents}_{it} + \delta_{14} \text{Country}_i + \delta_{15} \text{Industry}_i + \delta_{16} \text{Year}_t + \mu_{it} + \eta_i \quad (1a)$$

$$\text{Omission}_{it} = \delta_0 + \delta_1 \text{FemD}_{it} + \delta_2 \text{EnvPerf}_{it} + \delta_3 \text{BoarD}_{it} + \delta_4 \text{Meet}_{it} + \delta_5 \text{InD}_{it} + \delta_6 \text{Duality}_{it} + \delta_7 \text{SustCom}_{it} + \delta_8 \text{Size}_{it} + \delta_9 \text{Lev}_{it} + \delta_{10} \text{ROA}_{it} + \delta_{11} \text{Investment}_{it} + \delta_{12} \text{Investors}_{it} + \delta_{13} \text{DisrupEvents}_{it} + \delta_{14} \text{Country}_i + \delta_{15} \text{Industry}_i + \delta_{16} \text{Year}_t + \mu_{it} + \eta_i \quad (1b)$$

To test Hypothesis 2 and assess whether a critical mass of female directors is necessary to influence the disclosure of information

on decarbonization strategies, we replace the variable *FemD* with alternative measures of BGD that reflect different levels of female representation on the board. To do this, we construct Model 2.

$$\text{DescInf}_{it} = \alpha_0 + \alpha_1 \text{Skewed}_{it} + \alpha_2 \text{Tilted}_{it} + \alpha_3 \text{Balanced}_{it} + \alpha_4 \text{EnvPerf}_{it} + \alpha_5 \text{BoarD}_{it} + \alpha_6 \text{Meet}_{it} + \alpha_7 \text{InD}_{it} + \alpha_8 \text{Duality}_{it} + \alpha_9 \text{SustCom}_{it} + \alpha_{10} \text{Size}_{it} + \alpha_{11} \text{Lev}_{it} + \alpha_{12} \text{ROA}_{it} + \alpha_{13} \text{Investment}_{it} + \alpha_{14} \text{Investors}_{it} + \alpha_{15} \text{DisrupEvents}_{it} + \alpha_{16} \text{Country}_i + \alpha_{17} \text{Industry}_i + \alpha_{18} \text{Year}_t + \mu_{it} + \eta_i \quad (2a)$$

$$\text{DescInf}_{it} = \alpha_0 + \alpha_1 \text{OneWomen}_{it} + \alpha_2 \text{TwoWomen}_{it} + \alpha_3 \text{ThreeWomen}_{it} + \alpha_4 \text{EnvPerf}_{it} + \alpha_5 \text{BoarD}_{it} + \alpha_6 \text{Meet}_{it} + \alpha_7 \text{InD}_{it} + \alpha_8 \text{Duality}_{it} + \alpha_9 \text{SustCom}_{it} + \alpha_{10} \text{Size}_{it} + \alpha_{11} \text{Lev}_{it} + \alpha_{12} \text{ROA}_{it} + \alpha_{13} \text{Investment}_{it} + \alpha_{14} \text{Investors}_{it} + \alpha_{15} \text{DisrupEvents}_{it} + \alpha_{16} \text{Country}_i + \alpha_{17} \text{Industry}_i + \alpha_{18} \text{Year}_t + \mu_{it} + \eta_i \quad (2b)$$

$$\text{DescInf}_{it} = \alpha_0 + \alpha_1 \text{NoWomen}_{it} + \alpha_2 \text{EnvPerf}_{it} + \alpha_3 \text{BoarD}_{it} + \alpha_4 \text{Meet}_{it} + \alpha_5 \text{InD}_{it} + \alpha_6 \text{Duality}_{it} + \alpha_7 \text{SustCom}_{it} + \alpha_8 \text{Size}_{it} + \alpha_9 \text{Lev}_{it} + \alpha_{10} \text{ROA}_{it} + \alpha_{11} \text{Investment}_{it} + \alpha_{12} \text{Investors}_{it} + \alpha_{13} \text{DisrupEvents}_{it} + \alpha_{14} \text{Country}_i + \alpha_{15} \text{Industry}_i + \alpha_{16} \text{Year}_t + \mu_{it} + \eta_i \quad (2c)$$

$$\text{DescInf}_{it} = \alpha_0 + \alpha_1 \text{Women/FemD}_{it} + \alpha_2 \text{Women square/FemD square}_{it} + \alpha_3 \text{EnvPerf}_{it} + \alpha_4 \text{BoarD}_{it} + \alpha_5 \text{Meet}_{it} + \alpha_6 \text{InD}_{it} + \alpha_7 \text{Duality}_{it} + \alpha_8 \text{SustCom}_{it} + \alpha_9 \text{Size}_{it} + \alpha_{10} \text{Lev}_{it} + \alpha_{11} \text{ROA}_{it} + \alpha_{12} \text{Investment}_{it} + \alpha_{13} \text{Investors}_{it} + \alpha_{14} \text{DisrupEvents}_{it} + \alpha_{15} \text{Country}_i + \alpha_{16} \text{Industry}_i + \alpha_{17} \text{Year}_t + \mu_{it} + \eta_i \quad (2de)$$

Research hypotheses (Hypotheses 3a, 3b, and 3c) are tested using the Model 3 presented below, in which we interact the measure of BGD with *Scenario* to control for the moderating effect of the organizational and institutional environment:

$$\text{DescInf}_{it} = \beta_0 + \beta_1 \text{FemD}_{it} + \beta_2 \text{Scenario}_{it} + \beta_3 \text{FemD} * \text{Scenario}_{it} + \beta_4 \text{EnvPerf}_{it} + \beta_5 \text{BoarD}_{it} + \beta_6 \text{Meet}_{it} + \beta_7 \text{InD}_{it} + \beta_8 \text{Duality}_{it} + \beta_9 \text{SustCom}_{it} + \beta_{10} \text{Size}_{it} + \beta_{11} \text{Lev}_{it} + \beta_{12} \text{ROA}_{it} + \beta_{13} \text{Investment}_{it} + \beta_{14} \text{Investors}_{it} + \beta_{15} \text{DisrupEvents}_{it} + \beta_{16} \text{Country}_i + \beta_{17} \text{Industry}_i + \beta_{18} \text{Year}_t + \mu_{it} + \eta_i \quad (3)$$

In both models, *i* identifies the firm; *t* is the period; and δ and β are the coefficients to be calculated.

The proposed models are estimated using both ordinal and linear regressions for panel data, the latter using both fixed and random effects. The choice of ordinal methods in the basic models is guided by the ordinal quantitative nature of the dependent variable, *DescInf*. The use of linear regression methods allows us to guarantee that the results obtained with the previous methodology are robust to changes in the methodological specifications. However, Model 1b is estimated with probit and logit regressions for panel data, the latter with both fixed and random effects. The choice of these methods is guided by the dichotomous nature of the dependent variables.

To address potential causality issues, variables are instrumented using time lags, and the models incorporate firm-specific effects, where η denotes the unobservable heterogeneity and μ , the disturbance term.

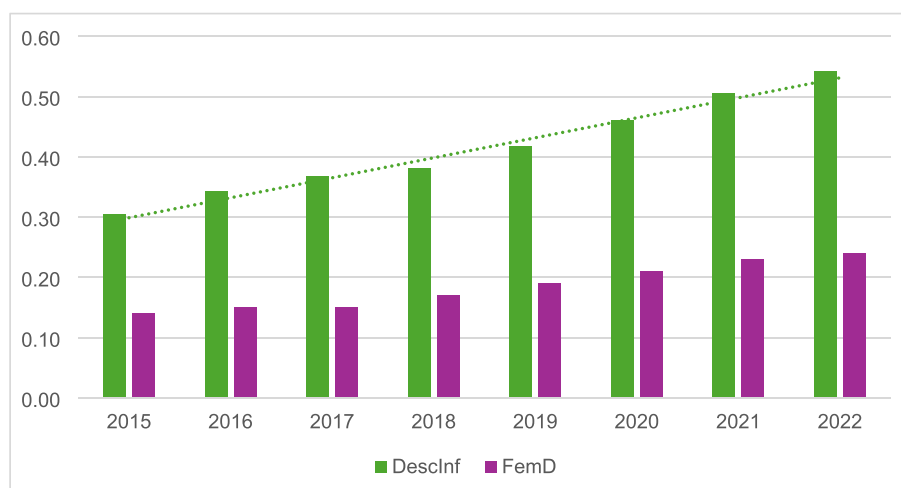


FIGURE 2 | Female directors and decarbonization disclosures.

4 | Empirical Findings

4.1 | Descriptive Statistics

Table 2 presents the descriptive statistics for the variables used to test the research hypotheses. Panel B highlights that the average proportion of female directors (FemD) is 19.17% and that large companies worldwide disclose an average of 43.49% of the nine decarbonization-related information items included in the DescInf variable. Panel A provides a detailed breakdown of the frequency of disclosure for each item, revealing significant opportunities for improvement, particularly regarding performance metrics, progress towards stated targets, and scenario analyses to bolster the resilience of corporate strategies.

Figure 2 illustrates the temporal evolution of the primary target variables under analysis, DescInf and FemD. A positive trend is observed for both variables across the study period, with a particularly pronounced increase in corporate transparency. Nevertheless, BGD remains far from a balanced composition, underscoring the persistent gap in achieving gender parity at a board level (García-Sánchez et al. 2024b).

Table 3 displays the bivariate correlations, confirming the absence of collinearity issues among the variables.

4.2 | Main Findings and Robustness Tests

4.2.1 | The Effect of BGD on Transparency of Decarbonization-Related Information

Table 4 reports the results of Model 1a, estimated using ordinal regression (Column 1) and linear regression methods (Columns 2 and 3, with fixed and random effects, respectively) for panel data. The inclusion of multiple methodological specifications strengthens the robustness of the findings.

Results from the ordinal regression in Column 1 reveal a positive effect of FemD on DescInf ($\delta_1 = 0.0198$), significant

at a 99% confidence level. This positive association is further confirmed in the fixed-effect and random-effects linear regressions (Columns 2 and 3, respectively). This results support Hypothesis 1, indicating that a higher representation of women on boards is positively associated with a more comprehensive disclosure of corporate decarbonization policies. This finding aligns with prior evidence obtained by Liao, Luo, and Tang (2015), Ben-Amar, Chang, and McIlkenny (2017), Charumathi and Rahman (2019), Hollindale et al. (2019), Ooi et al. (2019), Al-Qahtani and Elgharbawy (2020), and Tingbani et al. (2020), all of whom report a similar positive link between BGD and climate-related disclosures.

Regarding control variables, the analysis shows that larger, less indebted, and less profitable companies with stronger environmental performance are more inclined to report on decarbonization strategies. Disclosure is also positively correlated with larger, more active boards that have a greater proportion of executive members and a dedicated sustainability committee. Additionally, the influence of institutional initiatives in the 2020–2022 period on environmental reporting is evident. These findings are consistent with prior research on environmental disclosure (Al-Shaer and Zaman 2016; García-Sánchez et al. 2023a) and climate change reporting (Amran, Periasamy, and Zulkafli 2014; Liao, Luo, and Tang 2015; Ben-Amar, Chang, and McIlkenny 2017; Ararat and Sayedy 2019; Charumathi and Rahman 2019; Hollindale et al. 2019).

To further ensure the robustness of our findings, Table 5 presents the results based on alternative designs of the dependent variable and econometric methods suited to its nature. Specifically, we used the categorical variable Omission. Estimating Model 1b with this dummy variable allows us to examine the factors driving a company's choice to withhold information on its climate impact and progress towards emission reduction targets. At a 99% confidence level, the results indicate that the proportion of female directors has a negative effect ($\delta_1 = -0.0193$) on the likelihood of withholding such information. This further confirms that female board presence encourages the disclosure of more relevant information on corporate decarbonization efforts.

TABLE 3 | Bivariate correlations.

		1	2	3	4	5	6	7	8
1	DescInf	1							
2	FemD	0.23***	1						
3	SustInd	0.13***	-0.04***	1					
4	EU	0.23***	0.30***	0.00	1				
5	Oppo	0.42***	0.05***	0.13***	0.09***	1			
6	EnvPerf	0.81***	0.33***	0.01***	0.24***	0.37***	1		
7	BoarD	0.30***	0.05***	0.00	0.14***	0.18***	0.29***	1	
8	Meet	0.10***	0.02***	0.00	0.06***	0.07***	0.09***	0.02***	1
9	InD	-0.09***	0.33***	-0.08***	-0.06***	-0.09***	0.09***	-0.16***	-0.11***
10	Duality	-0.07***	-0.01*	-0.01**	-0.07***	-0.02***	-0.06***	0.02***	-0.12***
11	SustCom	0.77***	0.18***	0.12***	0.14***	0.30***	0.62***	0.24***	0.07***
12	Size	0.45***	0.07***	-0.02***	0.10***	0.25***	0.50***	0.49***	0.12***
13	Lev	0.04***	0.05***	0.00	-0.01***	0.03***	0.05***	0.01***	-0.02***
14	ROA	0.01**	0.01*	-0.01	0.00	0.00	0.01***	0.01	-0.02***
15	Investment	0.44***	0.10***	0.03***	0.12***	0.24***	0.49***	0.46***	0.07***
16	Investors	0.01	-0.02***	0.03***	0.03***	0.02***	-0.01**	-0.01	0.01
17	DisrupEvents	0.20***	0.21***	-0.01**	0.01***	0.08***	0.20***	-0.03***	0.07***
		9	10	11	12	13	14	15	16
9	InD	1							
10	Duality	0.15***	1						
11	SustCom	-0.08***	-0.08***	1					
12	Size	-0.09***	0.01	0.36***	1				
13	Lev	0.14***	0.08***	0.04***	0.09***	1			
14	ROA	0.01	0.00	0.01*	0.03***	0.01	1		
15	Investment	-0.06***	0.10***	0.32***	0.72***	0.05***	0.01	1	
16	Investors	-0.08***	-0.03***	0.01	0.00	-0.02***	0.00	-0.01**	1
17	DisrupEvents	0.07***	-0.01	0.18***	0.02***	-0.01**	0.00	0.00	0.00

*** $p < 0.01$.** $p < 0.05$.* $p < 0.1$.

4.2.2 | Critical Mass

Regarding Hypothesis 2, Table 6 presents the effects of varying levels of female board representation on board with respect to firms' disclosure practices on decarbonization strategies. Overall, the results suggest that the positive impact of female directors on transparency does not hinge on achieving a critical mass of women on the board; rather, each female director appears to bring a unique value, enhancing the firm's commitment to decarbonization transparency. Consequently, Hypothesis 2—which posits that the association between female board presence and increased transparency in decarbonization disclosures depends on reaching a critical mass—cannot be accepted.

The first column of Table 6 reflects the effect of female representation across the three board types defined by Seebeck and Vetter (2022) and García-Sánchez et al. (2023b) based on gender composition: Skewed, Tilted, and Balanced. The results show that female presence across all board types positively impacts the DescInf variable at a 99% confidence level. Notably, higher female representation correlated with stronger impacts: A presence below 20% has an impact of 0.210, between 20% and 40% it reaches 0.477, and at 40%–60% the impact rises to 0.571.

The fourth column, a model incorporating the number of female directors and its square, reveals linear relationship with no specific threshold altering the sign of the impact. This evidence contrasts with findings by Enciso-Alfaro and

TABLE 4 | Basic and robust results for Hypothesis 1: board gender diversity.

	Ologit regression	Linear regression	
	Coeff.	FE coeff.	RE coeff.
	(Std. error)	(Std. error)	(Std. error)
FemD	0.0198*** (0.00182)	0.00816*** (0.000811)	0.00621*** (0.000673)
EnvPerf	0.120*** (0.00181)	0.0425*** (0.000776)	0.0517*** (0.000625)
Board	0.0555*** (0.00948)	0.00413 (0.00384)	0.0182*** (0.00324)
Meet	0.0118*** (0.00359)	0.00438*** (0.00153)	0.00484*** (0.00137)
InD	−0.0174*** (0.00134)	−0.00244*** (0.000708)	−0.00787*** (0.000485)
Duality	−0.0625 (0.0526)	0.0556** (0.0243)	0.00157 (0.0194)
SustCom	2.328*** (0.0525)	0.879*** (0.0215)	1.152*** (0.0193)
Size	0.573*** (0.0289)	0.168*** (0.0200)	0.182*** (0.00981)
Lev	−0.275** (0.129)	−0.0660 (0.0576)	−0.123*** (0.0457)
ROA	−0.311*** (0.101)	−0.105*** (0.0364)	−0.0782** (0.0328)
Investment	0.0405 (0.0259)	0.0115 (0.0182)	0.0178* (0.00910)
Investors	−0.00510 (0.00610)	0.00669 (0.0914)	0.000309 (0.00169)
DisrupEvents	1.080*** (0.0300)	0.504*** (0.0123)	0.404*** (0.0110)
Country	Yes		Yes
Industry	Yes		Yes
Year	Yes		Yes
Log likelihood	−36674.38***		
R-squared		0.72***	0.73***

*** $p < 0.01$.** $p < 0.05$.* $p < 0.1$.

García-Sánchez (2024), who observed that for circular economy disclosures, female directors' influence depended on a balanced or near-balanced board composition.

Column 2 of Table 6, inspired by Atif et al. (2021), includes dummy variables for boards with one female director

(OneWomen), two female directors (TwoWomen), and three or more female directors (3+Women). The results confirm that the presence of female directors, regardless of their number, positively influences the likelihood of enhanced decarbonization strategy disclosure, which is consistent with Liao, Luo, and Tang (2015) but diverges from the findings of Ben-Amar, Chang,

TABLE 5 | Corrective effect of board gender diversity in omitting information on progress in reducing emissions.

	Probit regression	Logistic regression	
	Coeff.	FE coeff.	RE coeff.
	(Std. error)	(Std. error)	(Std. error)
FemD	−0.0193*** (0.00212)	−0.0667*** (0.00631)	−0.0350*** (0.00385)
EnvPerf	−0.0732*** (0.00211)	−0.127*** (0.00630)	−0.135*** (0.00390)
Board	−0.0223** (0.0101)	0.00967 (0.0308)	−0.0379** (0.0184)
Meet	−0.0226*** (0.00438)	−0.0471*** (0.0112)	−0.0412*** (0.00793)
IND	0.0101*** (0.00139)	−0.0157*** (0.00532)	0.0185*** (0.00251)
Duality	−0.0203 (0.0578)	−0.140 (0.169)	−0.0406 (0.105)
SustCom	−0.756*** (0.0561)	−0.695*** (0.150)	−1.411*** (0.104)
Size	−0.453*** (0.0300)	−1.872*** (0.167)	−0.828*** (0.0543)
Lev	−0.193 (0.141)	0.156 (0.465)	−0.358 (0.255)
ROA	−0.708*** (0.220)	−0.138 (0.651)	−1.264*** (0.401)
Investment	0.172*** (0.0257)	−0.00547 (0.121)	0.319*** (0.0463)
Investors	−0.00574 (0.00492)	0.748 (9.869)	−0.0104 (0.00841)
DisrupEvents	−1.169*** (0.0370)	−2.247*** (0.0927)	−2.153*** (0.0709)
Country	Yes		Yes
Industry	Yes		Yes
Year	Yes		Yes
Log likelihood	−9409.22***	−9381.81***	−9381.81***

*** $p < 0.01$.** $p < 0.05$.* $p < 0.1$.

and McIlkenny (2017) and Abbasi et al. (2024) for UK firms, who identified a threshold of at least two women on the board and audit committee, respectively, to influence climate-related disclosure. The results also contradict those of Charumathi and Rahman (2019) and Hollindale et al. (2019), who found that a single female director did not affect climate-related disclosure in Indian and Australian firms, respectively.

Our results confirm a stronger impact with increased female representation, showing an effect size of 0.866 when three or more seats are occupied by women, consistent with Atif et al. (2021). The model in the fifth column, incorporating the proportion of female directors and its square, confirms linear relationship with no proportion threshold altering the direction of the effect.

TABLE 6 | Basic and robust results for Hypothesis 2: Critical mass.

	Ologit regression				
	Coeff. (Std. error)	Coeff. (Std. error)	Coeff. (Std. error)	Coeff. (Std. error)	Coeff. (Std. error)
Skewed	0.210*** (0.0541)				
Tilted	0.477*** (0.0612)				
Balanced	0.571*** (0.0838)				
OneWomen		0.263*** (0.0549)			
TwoWomen		0.447*** (0.0649)			
3 + Women		0.866*** (0.0763)			
NoWomen			−0.297*** (0.0540)		
Women				0.264*** (0.0222)	
Women square				−0.0109 (0.0151)	
FemD					0.0253*** (0.00401)
FemD square					−0.000119 (7.72e-05)
EnvPerf	0.121*** (0.00181)	0.120*** (0.00181)	0.123*** (0.00180)	0.120*** (0.00181)	0.120*** (0.00181)
BoarD	0.0529*** (0.00959)	0.0254*** (0.00984)	0.0530*** (0.00956)	0.0308*** (0.0103)	0.0539*** (0.00954)
Meet	0.0123*** (0.00359)	0.0124*** (0.00359)	0.0129*** (0.00358)	0.0124*** (0.00359)	0.0119*** (0.00359)
InD	−0.0167*** (0.00134)	−0.0177*** (0.00135)	−0.0157*** (0.00133)	−0.0174*** (0.00134)	−0.0176*** (0.00135)
Duality	−0.0665 (0.0525)	−0.0668 (0.0526)	−0.0673 (0.0525)	−0.0693 (0.0526)	−0.0629 (0.0526)
SustCom	2.327*** (0.0525)	2.327*** (0.0525)	2.330*** (0.0524)	2.326*** (0.0525)	2.327*** (0.0525)
Size	0.569*** (0.0288)	0.571*** (0.0289)	0.564*** (0.0288)	0.568*** (0.0289)	0.574*** (0.0289)

(Continues)

TABLE 6 | (Continued)

	Ologit regression				
	Coeff. (Std. error)	Coeff. (Std. error)	Coeff. (Std. error)	Coeff. (Std. error)	Coeff. (Std. error)
Lev	−0.283** (0.129)	−0.281** (0.129)	−0.287** (0.129)	−0.280** (0.129)	−0.279** (0.129)
ROA	−0.314*** (0.101)	−0.332*** (0.100)	−0.307*** (0.101)	−0.323*** (0.100)	−0.316*** (0.101)
Investment	0.0425* (0.0259)	0.0392 (0.0259)	0.0446* (0.0258)	0.0360 (0.0259)	0.0400 (0.0259)
Investors	−0.00458 (0.00603)	−0.00491 (0.00607)	−0.00472 (0.00600)	−0.00505 (0.00609)	−0.00499 (0.00608)
DisrupEvents	1.105*** (0.0298)	1.081*** (0.0299)	1.132*** (0.0295)	1.082*** (0.0300)	1.078*** (0.0300)
Country	Yes	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Log likelihood	−36699.68***	−36665.19***	−36722.25***	−36663.28***	−36673.20***

*** $p < 0.01$.** $p < 0.05$.* $p < 0.1$.

Additionally, the results in Column 3 for the variable NoWomen show that all-male boards do not prioritize reporting on decarbonization strategies. The negative effect of NoWomen ($\delta_1 = -0.297$) is statistically significant at the 99% confidence level.

Figure 3 provides a visual representation of these relationships, illustrating the evolution of female board presence by both numbers and proportions over the period analyzed.

4.2.3 | The Moderating Role of Institutional and Business Environments

Table 7 reports the results of Model 3, estimated using ordinal regression for panel data. The first column of Table 7 illustrates the influence of the environmental sensitivity of industry sector. SustInd is significant at a 99% confidence level (0.886, $p < 0.01$), indicating that firms in high-impact sectors are more likely to disclose detailed information on decarbonization strategies, aligning with institutional theory. Regarding the industry's moderating effect on female directors' influence, the positive effect of the FemD*SustInd interaction (0.0205) implies that BGD plays a more substantial role in these sectors. This finding supports Hypothesis 3a. Female directors' influence on climate transparency is greater in environmentally sensitive industries, likely due to their responsiveness to external demands for sustainability transparency (Al-Qahtani and Elgharbay 2020). These results suggest that gender-diverse boards are more capable of meeting sustainability and transparency demands in environmentally

sensitive industries (Adams and Ferreira 2009; Harjoto, Laksmana, and Lee 2015).

In the second column, we assess the moderating effect of the EU institutional environment, where sustainability and transparency are prioritized. The EU variable has an impact of 1.303, which is significant at a 99% confidence level, underscoring that EU companies prioritize transparency on decarbonization strategies. This finding is consistent with institutional theory, which posits that organizations adapt to the social and regulatory pressures of their institutional context (Nicolò et al. 2022; García-Sánchez et al. 2023a). However, the FemD*EU interaction (-0.00819) suggests a corrective moderating effect where female directors' role in promoting transparency is slightly less pronounced, likely due to robust ESG regulations in the EU that lessen the need for individual influence. Consequently, Hypothesis 3b cannot be accepted. This result aligns with García-Sánchez et al. (2024b), indicating that strong institutional mandates reduce the individual impact of BGD, as companies already comply with high disclosure standards.

Finally, Column 3 reports the moderating effects of business opportunities linked to decarbonization. The Oppo variable displays a positive coefficient (0.621, $p < 0.01$), suggesting that firms identifying decarbonization-related opportunities are more transparent in this area, likely driven by both legitimacy considerations and economic incentives. However, the lack of a significant moderating effect on the influence of female directors on climate-related disclosures leads us to reject Hypothesis 3c. This finding implies that transparency around decarbonization-related strategies is already embedded as a strategic priority in

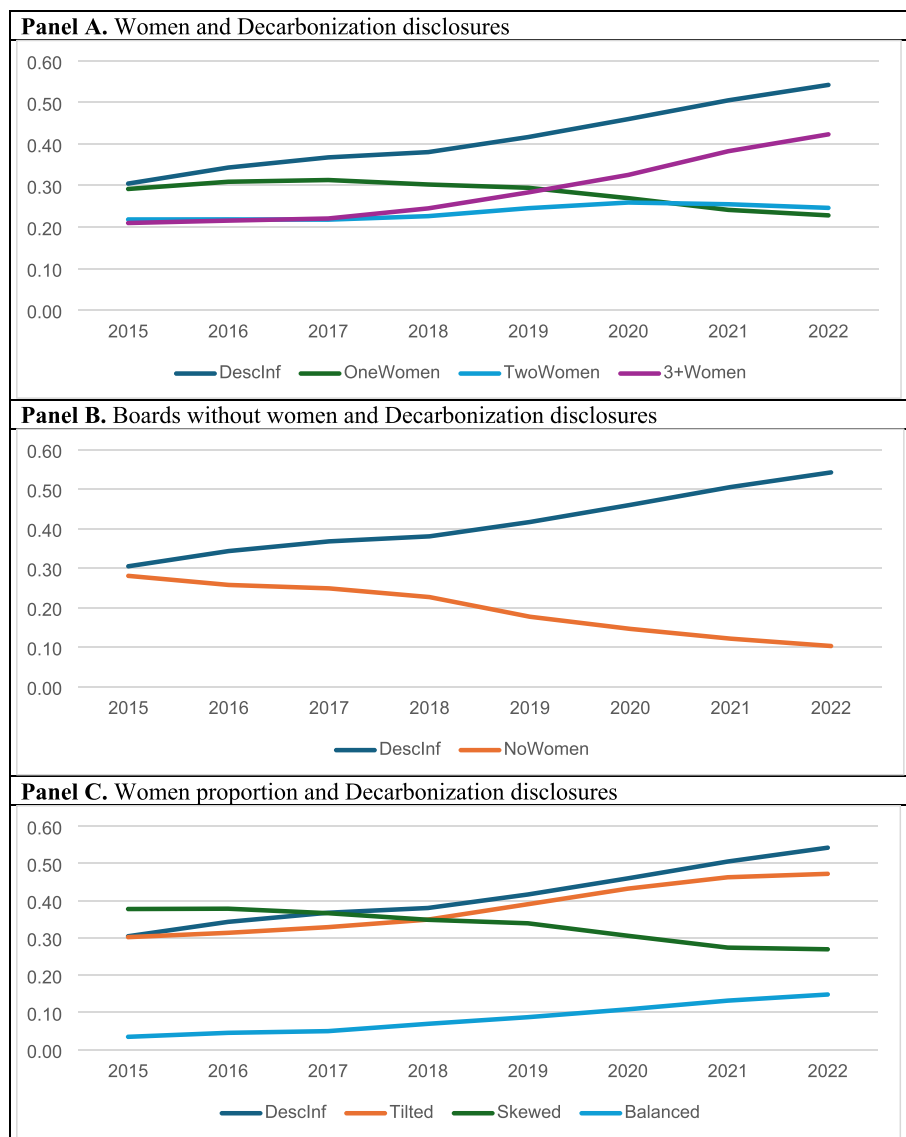


FIGURE 3 | Female presence and decarbonization disclosures.

firms pursuing these business opportunities, rendering it independent of gender diversity at the board level.

Finally, to illustrate the relationship between female directors and decarbonization-related disclosures, Table 8 presents statistics on these variables for each moderating scenario.

5 | Discussion and Conclusions

5.1 | Theoretical Implications

The findings support our hypothesis that the presence of women on corporate boards enhances transparency concerning decarbonization strategies, with clear theoretical implications across the three theories guiding this study. Regarding stakeholder theory, our results illustrate that gender-diverse boards are more attuned to stakeholder expectations—especially those relating to ESG—leading to prioritized disclosure of relevant, reliable information on climate change and decarbonization strategies. This contributes to the theory by demonstrating that enhanced

transparency among firms with greater BGD aligns closely with stakeholder expectations, fostering more accessible information that reinforces trust in the company's commitment to sustainability. From the perspective of the resource dependence theory, the results underscore gender diversity on boards as a resource that strengthens corporate strategic capacity. Female directors contribute distinctive perspective and heightened sensitivity to sustainability challenges, thereby affording the company a competitive advantage through superior management of the complexities surrounding decarbonization, alongside more effective communication of these strategies. In terms of gender identity theory, our findings suggest that female directors' increased responsiveness to sustainability issues may act as a catalyst for promoting more transparent decarbonization policies. We show that board members' gender identity can shape their priorities and perspectives on critical issues like sustainability and decarbonization, thus influencing boardroom dynamics and ultimately affecting transparency on decarbonization strategies.

Our results further suggest that this positive influence of female directors on decarbonization transparency is not contingent

TABLE 7 | Basic and robust results for Hypothesis 3: Institutional and organizational scenarios.

	Ologit regression		
	Coeff.	Coeff.	Coeff.
	(Std. error)	(Std. error)	(Std. error)
FemD	0.0162*** (0.00198)	0.0181*** (0.00196)	0.0201*** (0.00189)
SustInd	0.886*** (0.119)		
FemD*SustInd	0.0205*** (0.00398)		
EU		1.303*** (0.184)	
FemD*EU		−0.00819* (0.00486)	
Oppo			0.621*** (0.0998)
FemD*Oppo			−0.000129 (0.00378)
EnvPerf	0.121*** (0.00181)	0.119*** (0.00182)	0.118*** (0.00183)
Board	0.0536*** (0.00944)	0.0493*** (0.00948)	0.0540*** (0.00945)
Meet	0.0118*** (0.00359)	0.0121*** (0.00359)	0.0112*** (0.00359)
InD	−0.0174*** (0.00133)	−0.0173*** (0.00134)	−0.0168*** (0.00133)
Duality	−0.0703 (0.0523)	−0.0598 (0.0525)	−0.0677 (0.0524)
SustCom	2.302*** (0.0523)	2.339*** (0.0525)	2.327*** (0.0524)
Size	0.573*** (0.0286)	0.589*** (0.0289)	0.556*** (0.0287)
Lev	−0.261** (0.128)	−0.294** (0.129)	−0.278** (0.128)
ROA	−0.351*** (0.100)	−0.325*** (0.100)	−0.304*** (0.0998)
Investment	0.0355 (0.0256)	0.0279 (0.0258)	0.0428* (0.0257)
Investors	−0.00766 (0.00657)	−0.00697 (0.00640)	−0.00568 (0.00611)

(Continues)

TABLE 7 | (Continued)

	Ologit regression		
	Coeff.	Coeff.	Coeff.
	(Std. error)	(Std. error)	(Std. error)
DisrupEvents	1.080*** (0.0300)	1.100*** (0.0301)	1.065*** (0.0300)
Country	Yes	Yes	Yes
Industry	Yes	Yes	Yes
Year	Yes	Yes	Yes
Log likelihood	−36574.37***	−36634.05***	−36625.90***

*** $p < 0.01$.** $p < 0.05$.* $p < 0.1$.

TABLE 8 | Descriptive statistics by scenarios.

	Total	SustInd		EU		Opo	
		0	1	0	1	0	1
DescInf	43.49	40.72	48.35	40.27	55.26	37.61	67.33
FemD	19.17	18.75	17.04	16.89	29.32	18.27	19.69
Balanced	8.33	8.37	7.43	5.93	24.60	8.04	9.90
Tilted	36.92	37.74	33.28	34.92	50.49	36.43	39.59
Skewed	33.45	33.37	34.50	35.95	16.49	33.68	32.19
3 + Women	27.03	27.78	24.40	22.57	57.25	25.11	37.53
TwoWomen	23.57	23.80	21.79	24.03	20.46	23.91	21.70
OneWomen	28.84	28.65	29.64	30.85	15.21	29.90	23.04
NoWomen	20.56	19.77	24.17	22.54	7.08	21.07	17.73

upon a critical mass of women on the board, as even a single female director has a measurable impact, with the effect intensifying as more female directors join. This challenges the application of critical mass theory in this specific context, proposing instead that the visibility and influence of even one female directors on sustainability topics can bring transparency issues to the boards' attention.

Incorporating these findings with those of the institutional theory analysis, we observe that external context is also relevant to the relationship between BGD and transparency in decarbonization. In environmentally sensitive industries, the positive influence of BGD on transparency is amplified, aligning with the institutional theory assertion that organizations respond to contextual pressures to legitimize their practices. Conversely, in the EU environment, where stringent ESG regulations prevail, the universal push for transparency diminishes the need for gender diversity on boards as a determinant. For firms identifying business opportunities in decarbonization, transparency appears embedded within their strategic approach and shows no direct reliance on BGD, suggesting that when sustainability aligns closely with core strategy, the presence of female directors may not be essential. Collectively, these findings imply that pressures

for transparency within sustainability initiatives differentially condition the role of female directors in corporate disclosure on decarbonization strategies. These findings not only highlight the capacity of female directors to shape strategic decisions critical to sustainable development but also emphasizes the importance of integrating both internal organizational attributes and external institutional context when evaluating corporate sustainability practices.

5.2 | Practical Implications

In practical terms, this study offers valuable insights for managers, investors, regulators, and society. For corporate managers, the findings underscore the strategic value of BGD as a lever to enhance transparency, particularly on sensitive topics like sustainability and decarbonization, which in turn strengthens corporate reputation and legitimacy. For investors, this research suggests that firms with a female board representation are more likely to commit to transparent sustainability strategies, signaling potentially lower regulatory risk and better alignment with responsible investment criteria. For regulators, the findings advocate for promoting gender diversity on boards as an effective

means of advancing transparency in sustainability practices, a critical component aspect of the transition to sustainable and equitable economies.

From a societal perspective, the study highlights the broader societal benefits of fostering gender equality in the corporate sphere, underscoring the transformative role women can play in strategic decision-making relevant to sustainable development, particularly in sectors with high environmental impacts. In this way, our research advances theoretical insights into governance and sustainability while providing actionable pathways for meaningful change in corporate practice, with far-reaching positive implications for society.

5.3 | Limitations and Future Research Avenues

Whereas this study offers significant insights into the impact of BGD in promoting transparency around decarbonization strategies, several limitations must be acknowledged. First, although the sample includes a broad representation of large global companies, its focus on the largest corporations may limit the generalizability of the findings to smaller organizations, which may face distinct challenges and constraints in implementing decarbonization strategies. Second, this study employs a quantitative approach, which precludes an in-depth exploration of the internal board dynamics that may influence climate disclosures. Additionally, this study does not account for the demographic characteristics or specific expertise and knowledge of female directors, which may play a critical role in shaping their influence on decarbonization transparency. Finally, although the study examines a range of contextual factors, other elements—such as specific national policies or the influence of local stakeholders—could impact the relationship between BGD and climate transparency and have not been exhaustively addressed.

Building upon the limitations identified, several avenues for future research emerge. Expanding the analysis to include smaller firms could provide a more comprehensive understanding of the relationship between BGD and decarbonization transparency. Additionally, qualitative methodologies, such as in-depth interviews or case studies, could offer valuable insights into the internal dynamics of boards, revealing how gender diversity and other board characteristics shape strategic decision-making and transparency practices. Examining the demographic characteristics and specific expertise of female board members also represents a promising direction, as it could shed light on how these attributes contribute to board effectiveness in addressing climate-related issues. Furthermore, future research could explore the role of national policy frameworks and local stakeholder pressures in moderating the influence of BGD on sustainability transparency. Finally, longitudinal studies tracking changes in board composition and transparency practices over time could enhance our understanding of causal relationships and the long-term impact of gender-diverse boards on corporate sustainability.

5.4 | Concluding Remarks

Our findings reinforce the view that gender diversity in corporate boards is more than a corporate social responsibility objective; it

is a foundational element of improved governance and risk management, particularly in the face of climate-related challenges. Female directors' influence in shaping corporate culture and advocating for enhanced accountability and transparency appears to be a critical driver in adopting more sustainable and responsible practices. In a global landscape that increasingly demands corporate accountability, equity, and transparency, this study provides critical evidence and guidance for academics, regulators, investors, and managers seeking to build more responsible, sustainable, and equitable organizations.

Conflicts of Interest

The authors declare no conflicts of interest.

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